BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

Z(→ℓℓ) + b/c AK4 jets in Run 3

Summer22 DY→ℓℓ MC with AK4 jets: samples, cutflow, kinematics, and a comparison with the boosted AK8 selection

11 October 2026

Campaign and setup

Goal: measure σ(Z+b) / σ(Z+c) with an AK4 jet recoiling against Z→ee/μμ (switched from the boosted AK8 jet after the group meeting). First step: tune the selection on DY→ℓℓ MC.

MC CAMPAIGN

Summer22

Run3Summer22 = 2022 eras C–D (pre-EE)

FORMAT

NanoAODv12

130X MC conditions

ENERGY (√s)

13.6 TeV

Run 3 pp collisions

NORMALIZATION

7.98 fb^{-1}

2022 C–D golden JSON; brilcalc check pending

MC only, no data yet. Processed: 1 file per sample (test run), the same files as the AK8 test. Code: BTVNanoCommissioning, branch coffea_machine, workflow Zb_Zc_ratio (AK4, commit f8748a6); the AK8 selection runs as Zb_Zc_ratio_AK8.

MC samples: DY→ℓℓ, stitched in pT(ℓℓ)

```
/DYto2L-2Jets_MLL-50[_PTLL-X]_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/
  Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-vN/NANOAODSIM
```

Sample	LHE pT(ℓℓ) used	σ [pb]	DAS files	DAS events	Test: events used
Inclusive (MLL-50)	< 100 GeV	6346	525	93.8 M	197,672
PTLL-100	100–200 GeV	105.0	418	56.4 M	161,921
PTLL-200	200–400 GeV	11.24	335	44.7 M	91,186
PTLL-400	400–600 GeV	0.643	424	41.6 M	2,727
PTLL-600	> 600 GeV	0.0994	185	9.50 M	939

PTLL-X samples are inclusive above X, so each is kept to its own LHE pT(ℓℓ) slice. σ(PTLL-X) = 6346 pb × fraction of inclusive events with pT(ℓℓ) > X (measured on 27.7 M inclusive events). "Test: events used" = events of 1 file inside the slice. With no boosted-jet requirement the inclusive sample (pT(ℓℓ) < 100 GeV) gives 87% (Z→ee) / 89% (Z→μμ) of the scaled yield from only 11–13% of the raw selected events, so its large per-event weights limit the MC statistics.

Event selection, per channel

Z→ee / Z→μμ (unchanged)

- Trigger: HLT_Ele30_WPTight_Gsf / HLT_IsoMu24
- Leptons pT > 25 GeV, |η| < 2.4; leading pT > 35 GeV
- e: tight cut-based ID, IP cuts, EB–EE gap veto
- μ: tight ID, PF isolation < 0.15
- Opposite sign (OS)
- 71 < m$\ell\ell$ < 111 GeV
- PuppiMET < 50 GeV

Leading AK4 jet ("jet" step)

- AK4 PUPPI jets, ΔR > 0.4 from selected leptons
- Tight-lepton-veto jet ID (with NanoAODv12 jetId fix)
- pT > 30 GeV, |η| < 2.5; ≥ 1 such jet, leading one used
- No Z pT requirement

Weights and flavour

- genWeight × pileup × lepton ID/iso/reco SFs
- No trigger SF yet
- Jet flavour from hadronFlavour: b (5), c (4), light; no bb/cc split for AK4
- Also labelled by |partonFlavour|: b, c, uds, g, none (pages 21+, CMS recommendation on p. 21)
- Tagger: PNet AK4 B, CvL, CvB

Cutflow, Z→ee: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	197,672	32,660	9,802	9,745	9,396	8,872	2,347	26.5%
PTLL-100	161,921	46,448	13,156	13,037	12,032	10,642	10,214	96.0%
PTLL-200	91,186	31,202	11,594	11,431	10,398	8,584	8,466	98.6%
PTLL-400	2,727	1,013	458	447	396	287	286	99.7%
PTLL-600	939	229	101	100	82	54	54	100.0%

21,367 raw MC events pass the full Z→ee selection; the most (48%) come from PTLL-100. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT($\ell\ell$) > 100 GeV.

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · "all" = events in the sample's LHE pT($\ell\ell$) slice · 5

Cutflow, Z→μμ: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	197,672	42,062	17,685	17,685	17,054	16,139	4,159	25.8%
PTLL-100	161,921	48,592	20,606	20,604	19,043	16,788	16,142	96.2%
PTLL-200	91,186	31,717	16,868	16,865	15,258	12,453	12,298	98.8%
PTLL-400	2,727	1,048	659	659	592	407	405	99.5%
PTLL-600	939	319	199	199	178	102	102	100.0%

33,106 raw MC events pass the full Z→μμ selection; the most (49%) come from PTLL-100. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT($\ell\ell$) > 100 GeV.

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · "all" = events in the sample's LHE pT($\ell\ell$) slice · 6

Yields by AK4 jet flavour (7.98 fb^{-1})

Channel	light	c	b	total
Z→ee	343,532.5	42,631.0	16,844.4	403,007.9
fraction	0.852	0.106	0.042	
Z→μμ	683,956.5	65,108.4	35,637.3	784,702.2
fraction	0.872	0.083	0.045	

Still 1 file per sample

No negative yields now: 21,367 (Z→ee) and 33,106 (Z→μμ) raw events pass, vs ~1.4k / 2.1k with AK8. Single negative-weight amcatnlo events still leave jumps and empty bins in the tails of the plots. No b-tag requirement applied yet.

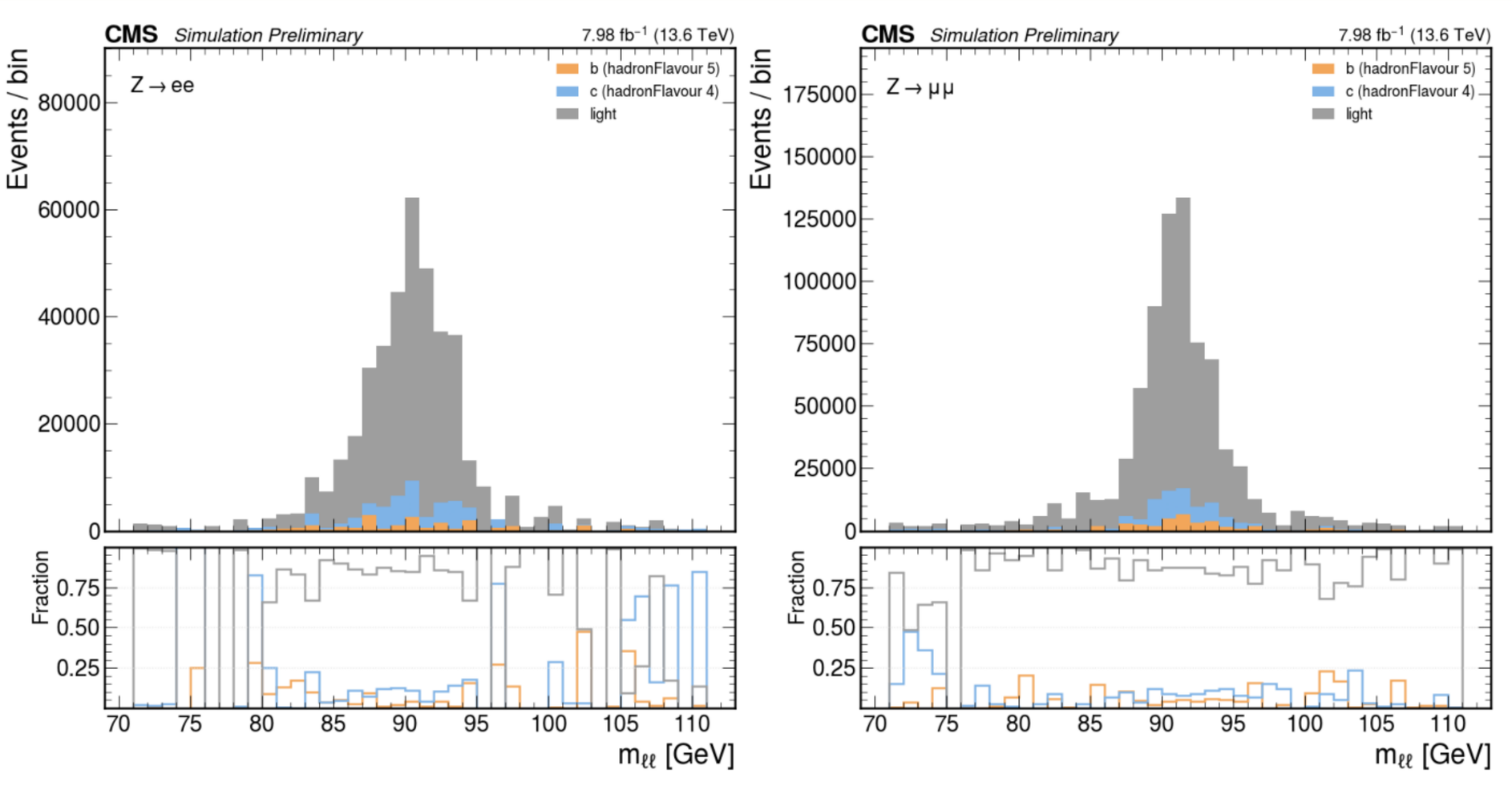
AK8 vs AK4: same files, same Z selection

	AK8 Z→ee	AK4 Z→ee	AK8 Z→μμ	AK4 Z→μμ
Raw events selected	1,440	21,367	2,121	33,106
Scaled yield	2,259.7	403,007.9	2,784.7	784,702.2
b fraction (b + bb)	0.043	0.042	−0.045	0.045
c fraction (c + cc)	0.118	0.106	0.244	0.083
light fraction	0.840	0.852	0.800	0.872
Jet requirement	pT > 200, mSD > 40	pT > 30	pT > 200, mSD > 40	pT > 30

AK8 flavour is from B/C hadron counts inside the jet (bb/b/cc/c); AK4 flavour is hadronFlavour. The AK4 selection keeps ~15× more events because it does not require a boosted Z. AK8 heavy-flavour fractions are unreliable at this statistics (negative Z→μμ b yield).

Dilepton mass

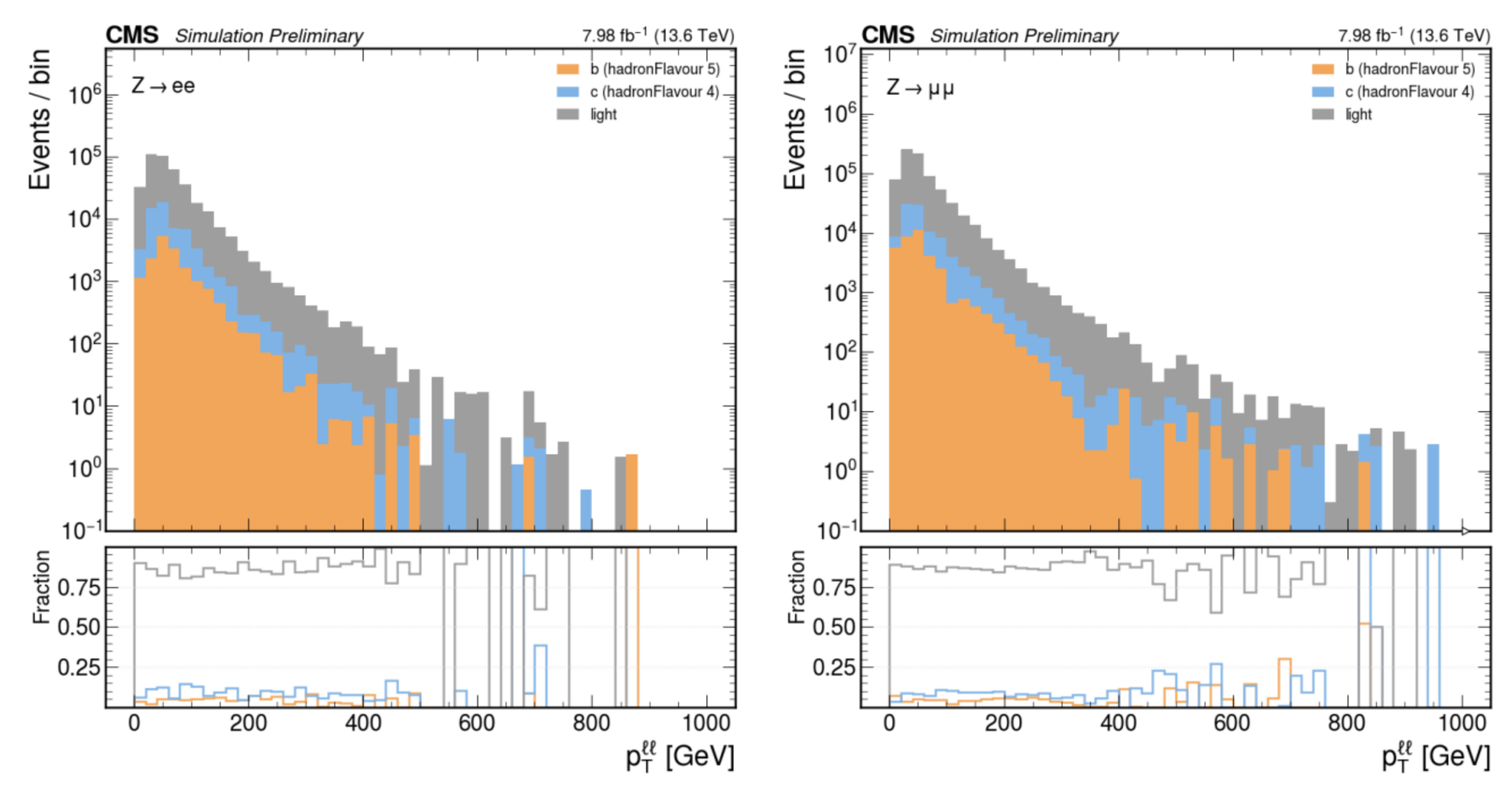
m(ℓℓ) after the full selection (71–111 GeV window): peaks at the Z mass in both channels. Linear y-axis.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 9

Dilepton pT

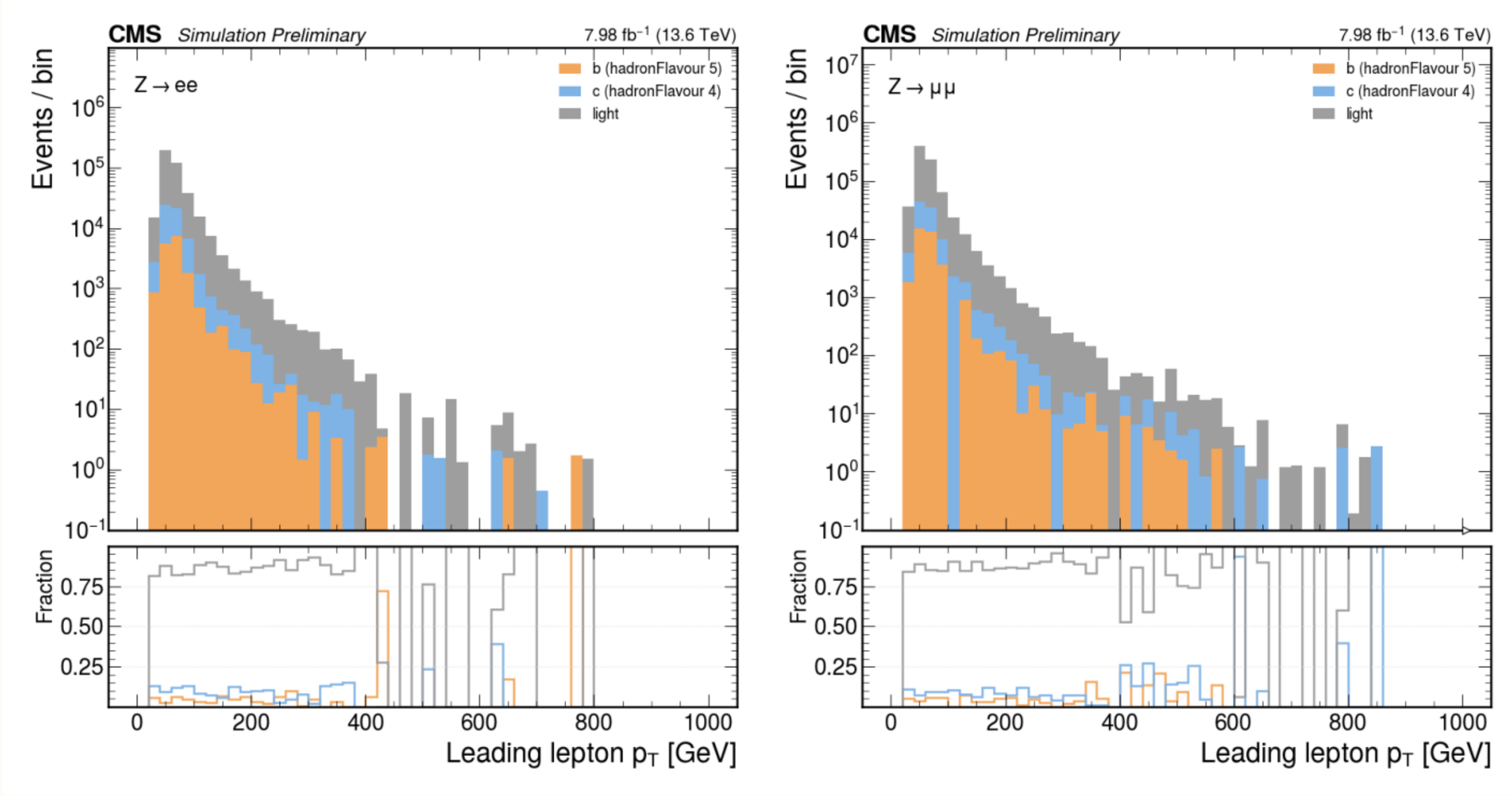
pT(ℓℓ) of the Z candidate after the full selection. No Z pT cut: dominated by low pT, unlike the boosted AK8 selection.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 10

Leading-lepton pT

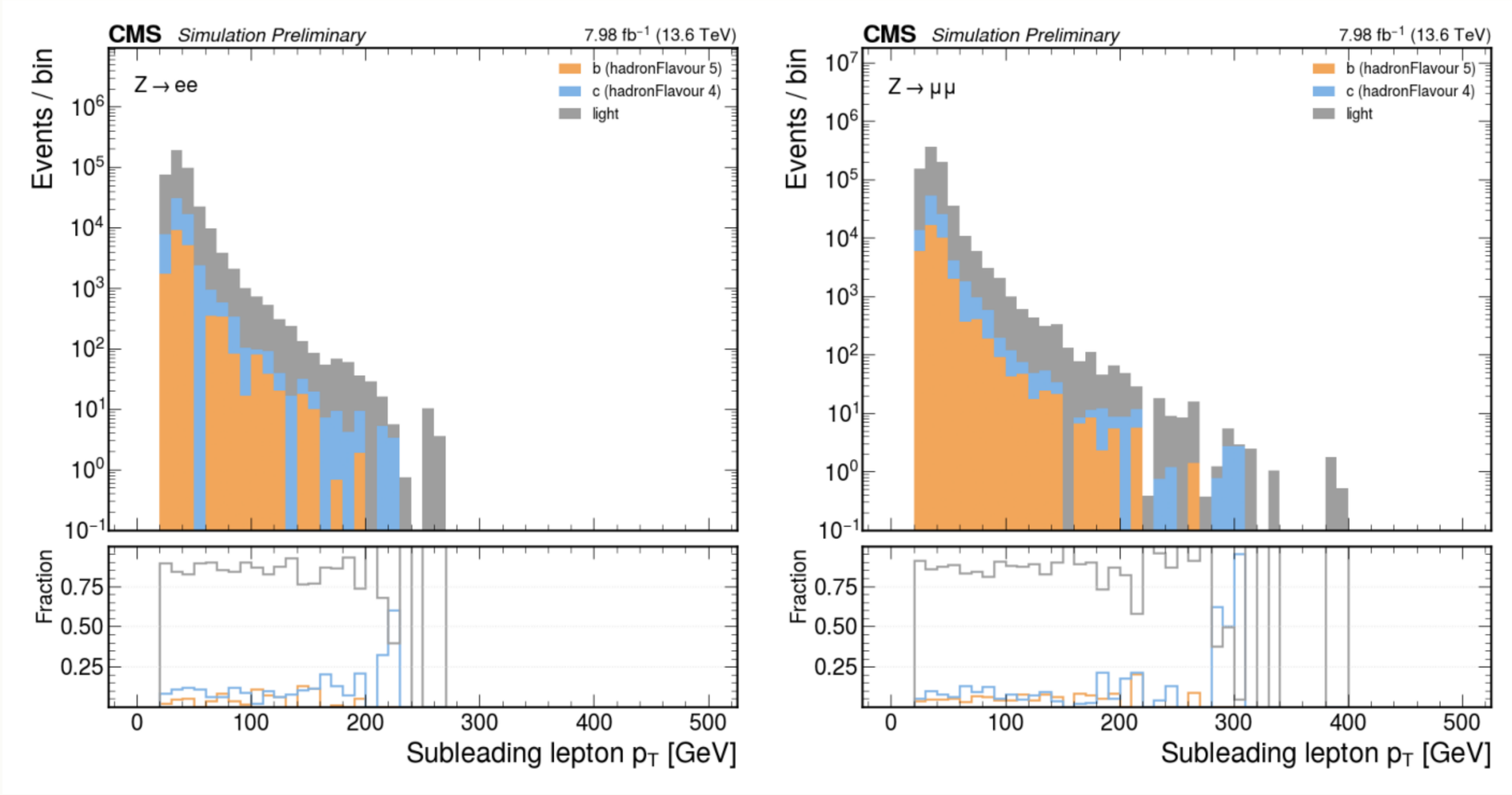
Leading lepton of the Z candidate, after the full selection (pT > 35 GeV required).



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 11

Subleading-lepton pT

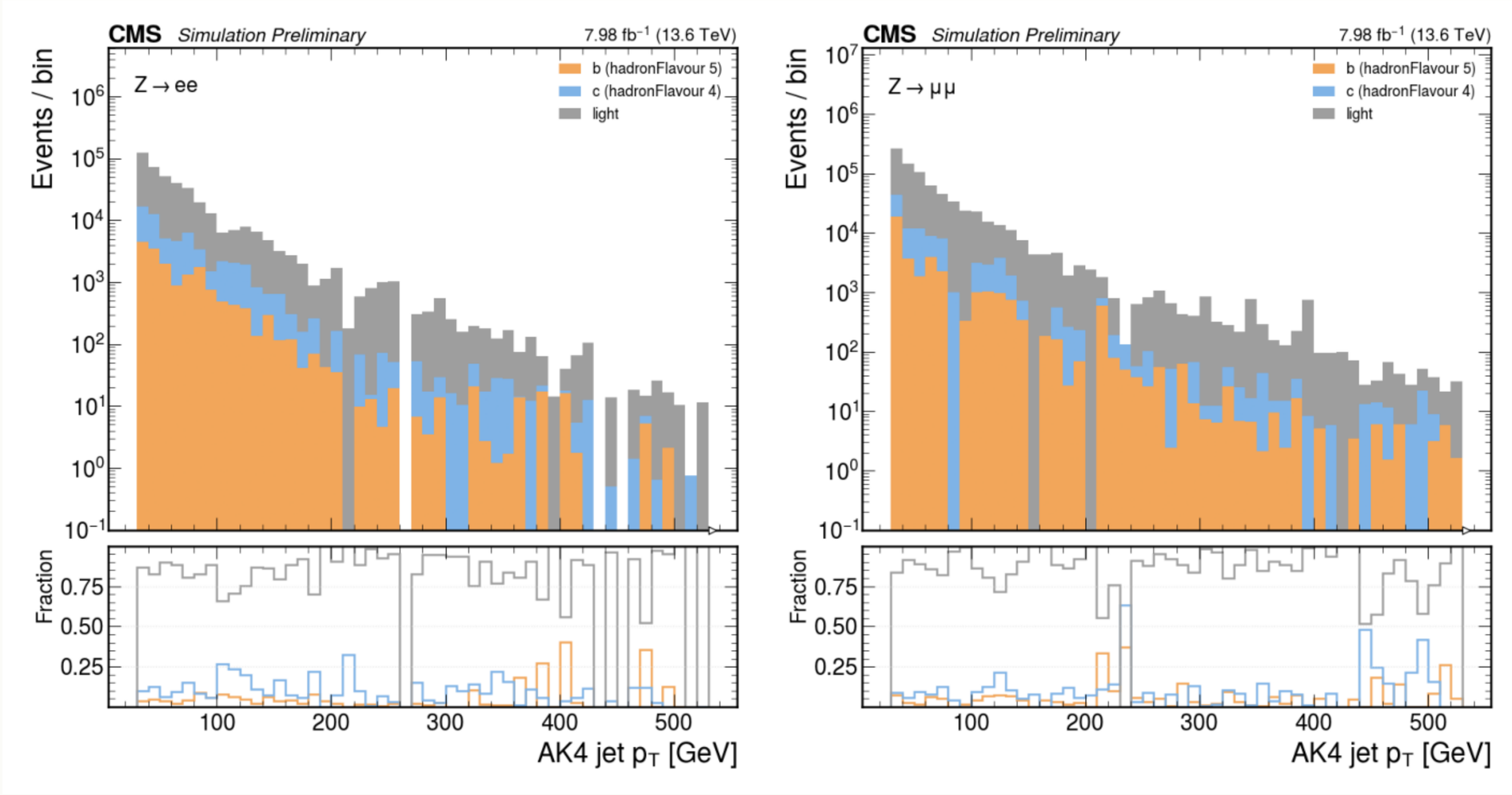
Subleading lepton of the Z candidate, after the full selection (pT > 25 GeV required).



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 12

Leading AK4 jet pT

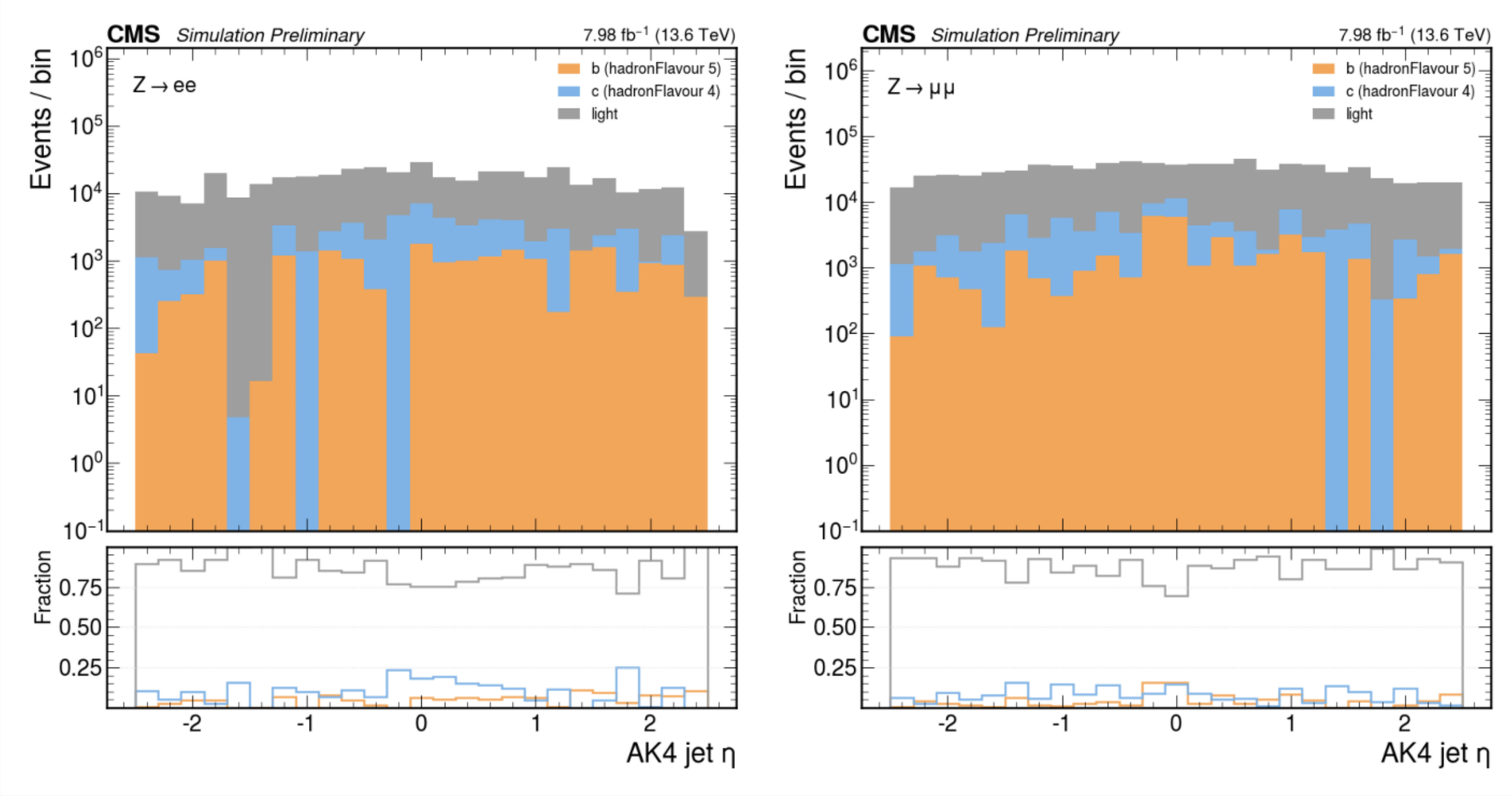
Leading lepton-cleaned AK4 jet with tight-lepton-veto ID (pT > 30 GeV required).



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 13

Leading AK4 jet η

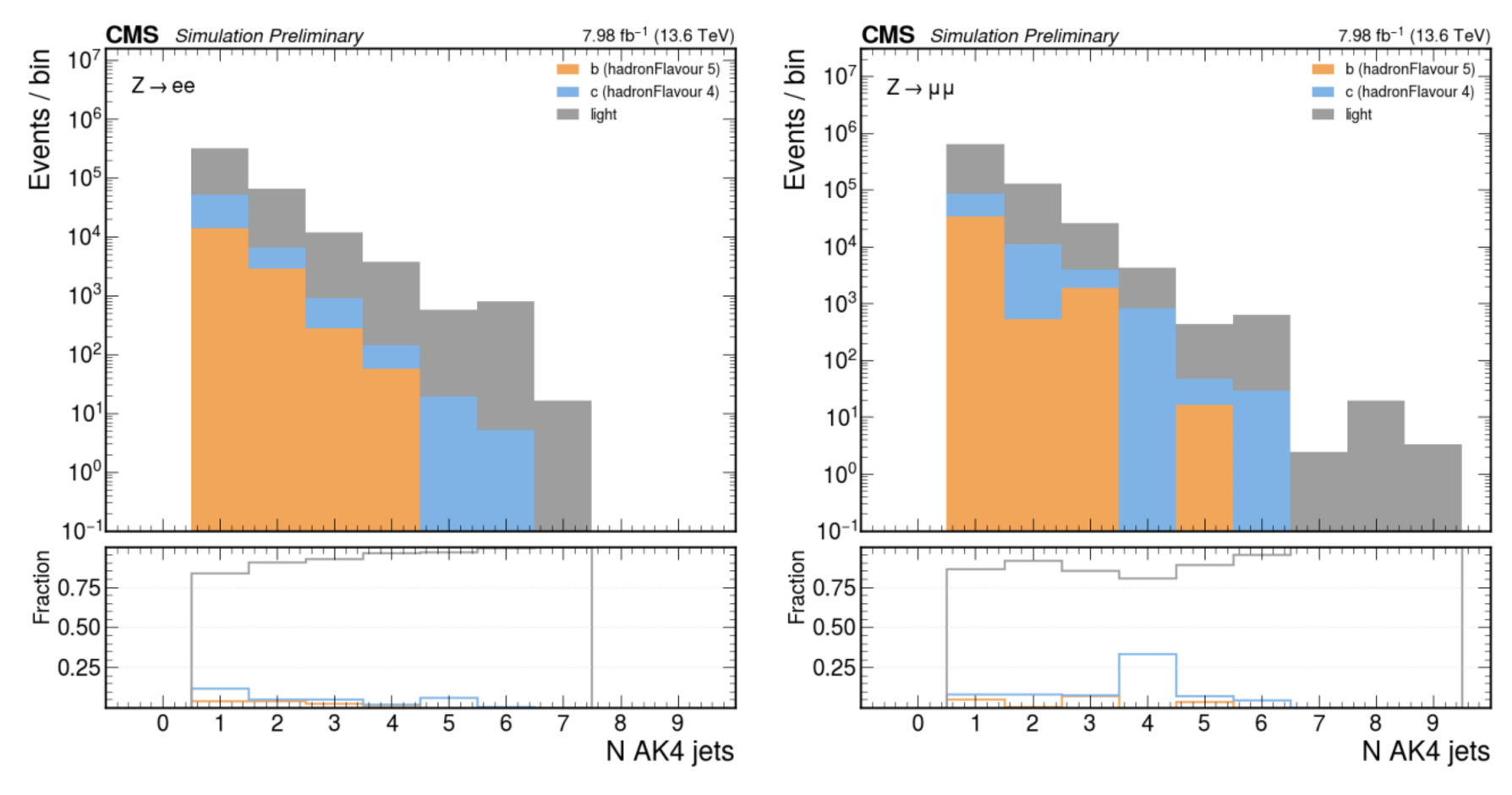
Pseudorapidity of the selected AK4 jet (|η| < 2.5 required).



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 14

AK4 jet multiplicity

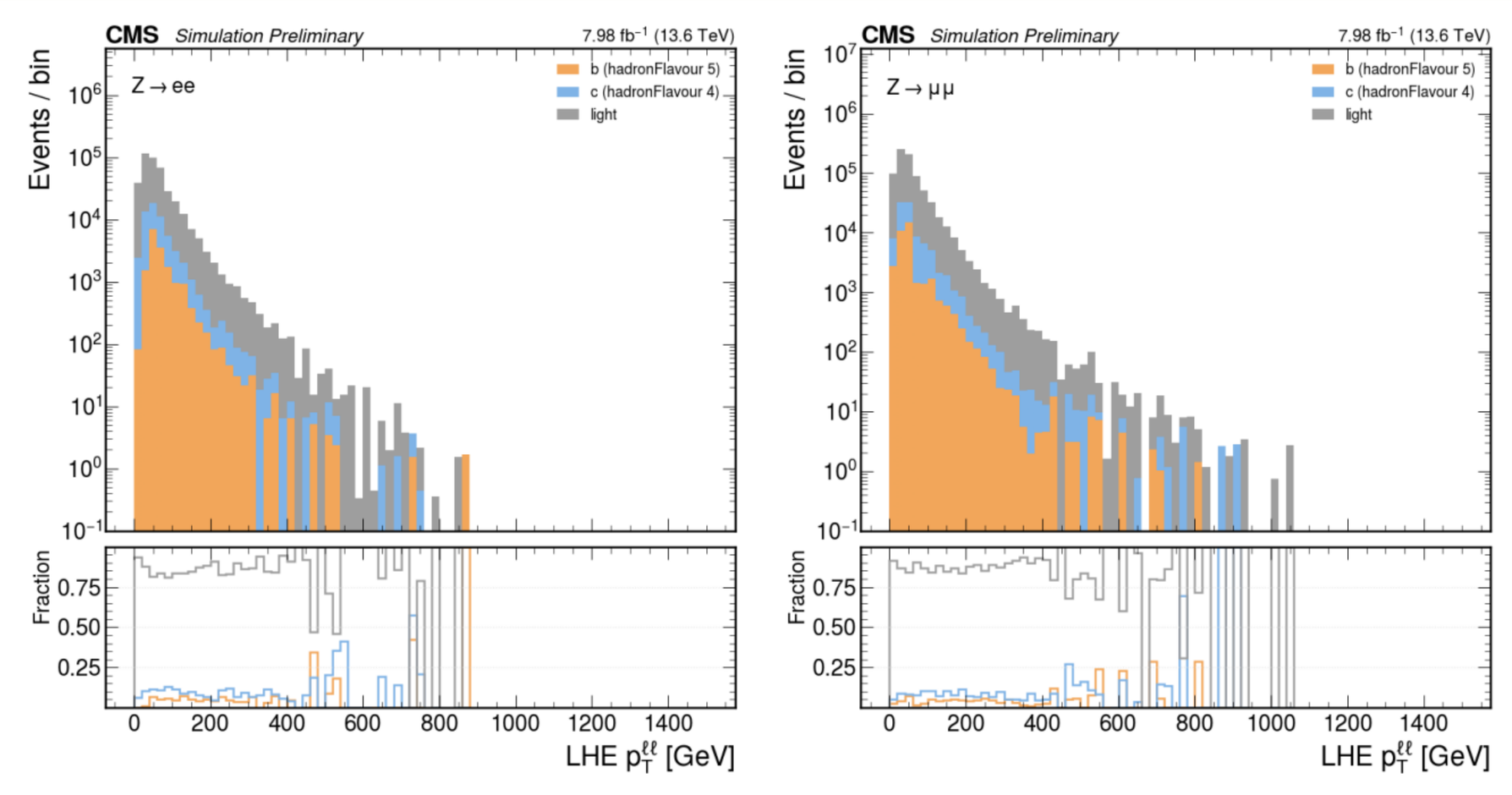
Number of selected AK4 jets (pT > 30 GeV, |η| < 2.5, ID, lepton-cleaned) in events passing the full selection.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 15

LHE pT($\ell\ell$)

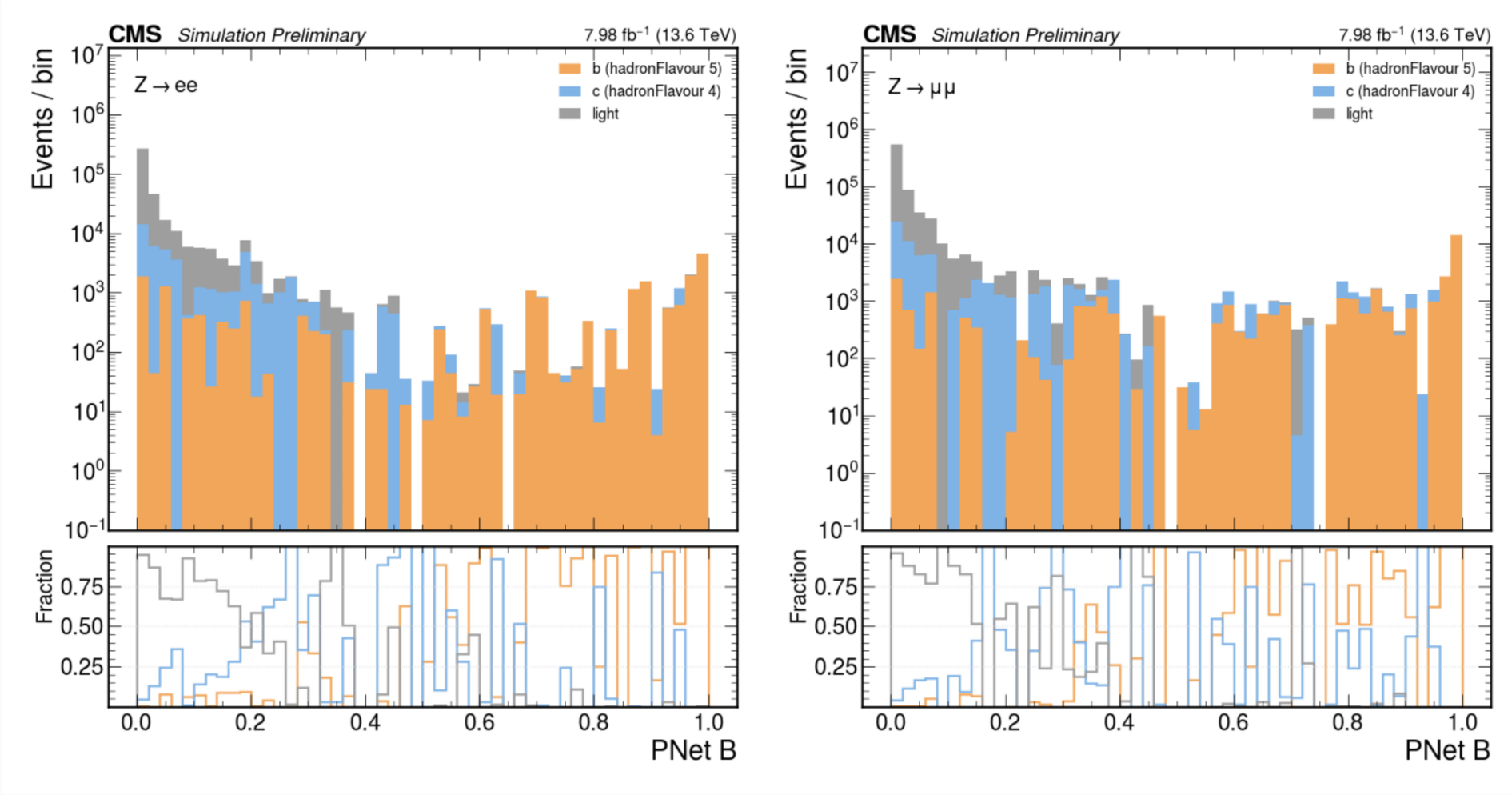
Generator-level pT($\ell\ell$) after the full selection: checks the sample stitching (slices at 100/200/400/600 GeV).



Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 16

Leading AK4 jet: PNet B

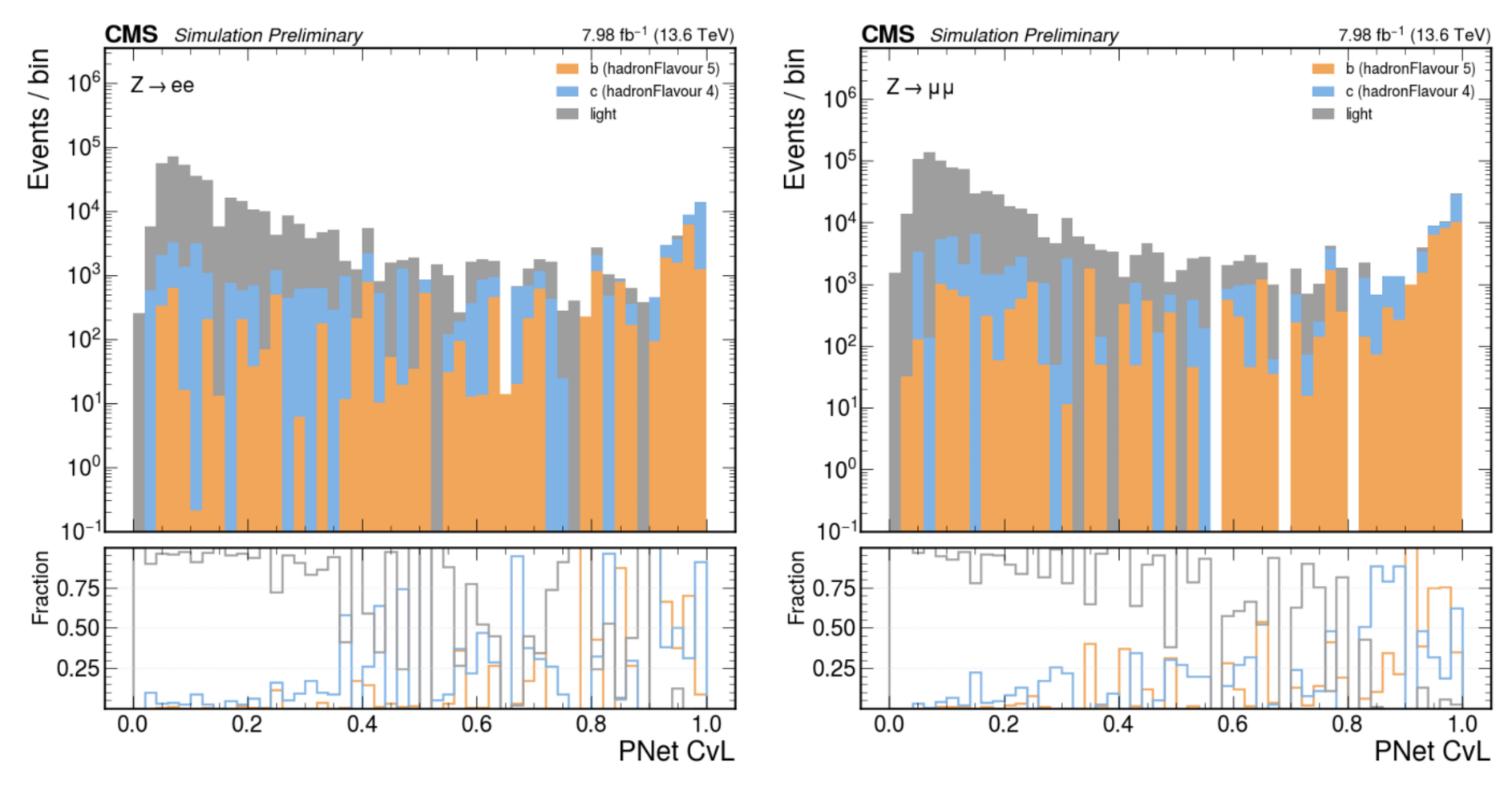
ParticleNet b-tag score of the selected jet; b jets pile up near 1, light jets near 0.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 17

Leading AK4 jet: PNet CvL

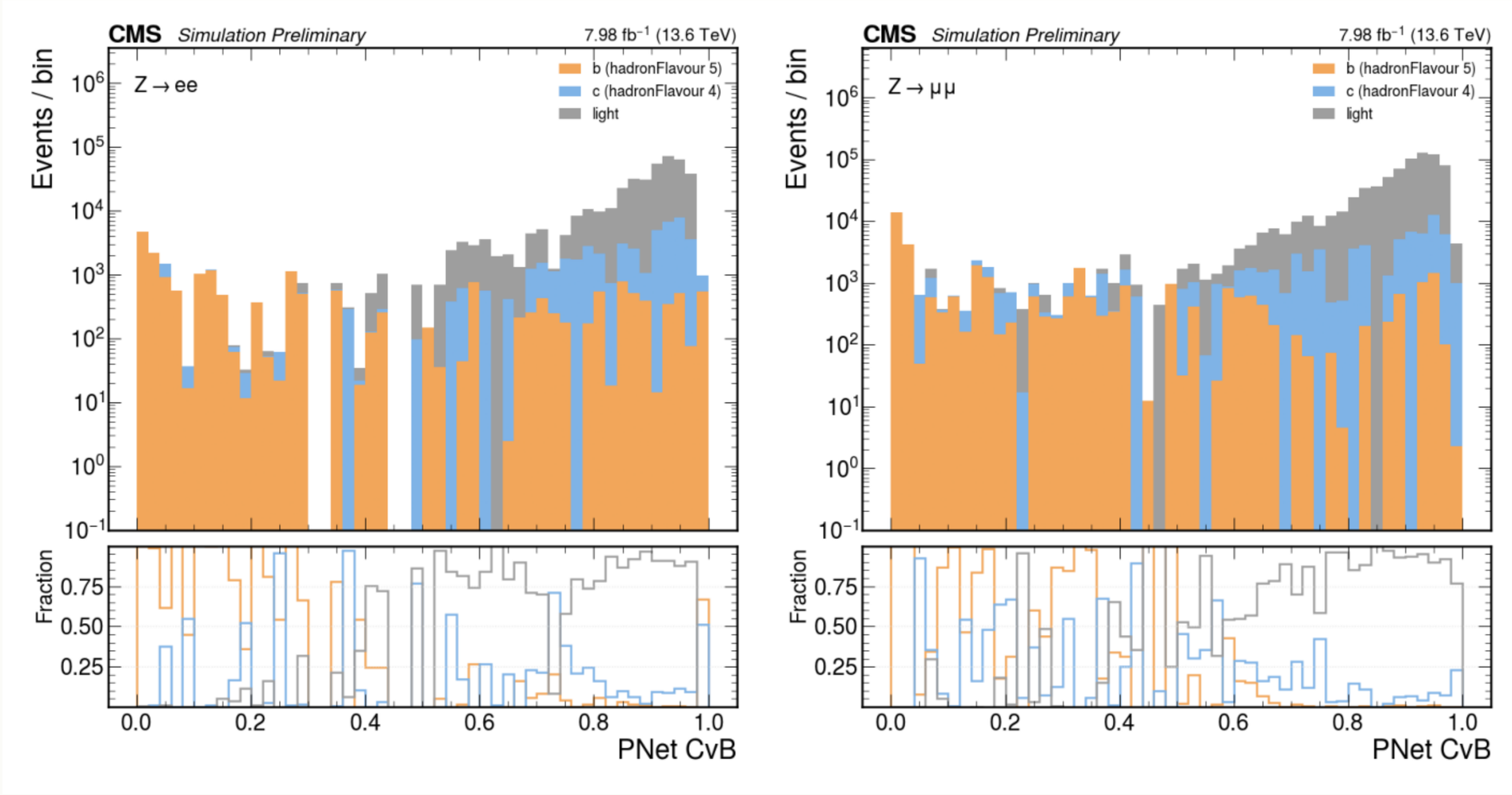
ParticleNet charm-vs-light score of the selected jet.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 18

Leading AK4 jet: PNet CvB

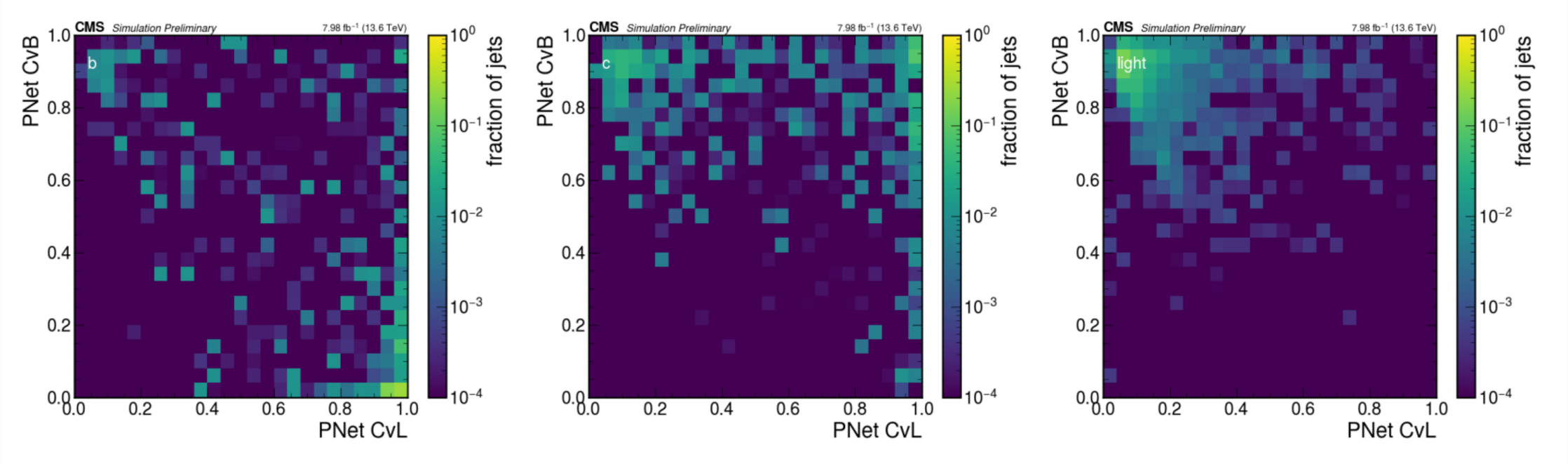
ParticleNet charm-vs-b score of the selected jet.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 19

PNet charm-tagging plane: CvL vs CvB

Selected AK4 jets per flavour, both channels summed, normalized to unit area. b jets sit at high CvL and low CvB, light jets at low CvL and high CvB. c jets are at high CvB, but a large part of them overlaps the light-jet corner (low CvL), with the rest at high CvL.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · Z→ee + Z→μμ · 20

Jet flavour: what CMS recommends

CMS recommendation (BTV)

- Use the hadron-based flavour (hadronFlavour), the CMSSW default
- Generator B and C hadrons are clustered into the jet with ~zero momentum, so the jet itself is unchanged
- b if the jet has a B hadron, c if it has a C hadron but no B, light otherwise
- All BTV b/c-tag scale factors are given for this definition
- partonFlavour: mainly to separate gluon from uds jets

What we do

- Take Jet_hadronFlavour from NanoAOD for the leading jet: 5 = b, 4 = c, rest = light
- No matching of our own
- No b/bb or c/cc split: NanoAOD doesn't store hadron counts for AK4 jets
- partonFlavour only as a cross-check: agrees for 99% of b jets and 92% of c jets

Conclusion: we are on the right track

Our flavour labels are exactly the ones CMS recommends, so the BTV scale factors can be applied as they are. The c/b ratio in our MC (2.5 for Z→ee, 1.8 for Z→μμ) is a bit higher than, but in the same ballpark as, the CMS 13 TeV measurement, 1.62 ± 0.15 (PRD 102, 032007). Not a like-for-like comparison yet: ours is MC truth for the leading jet, without tagging.

Reference: CMS TWiki, Jet Flavour Identification (MC Truth)

https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideBTagMCTools#Hadron_parton_based_jet_flavour

hadronFlavour vs partonFlavour

hadronFlavour (default)

- Ghost-matched B / C hadrons clustered into the jet
- b (5) if any B hadron, else c (4) if any C hadron, else 0
- BTV standard; b/c-tag SFs are measured for it

partonFlavour

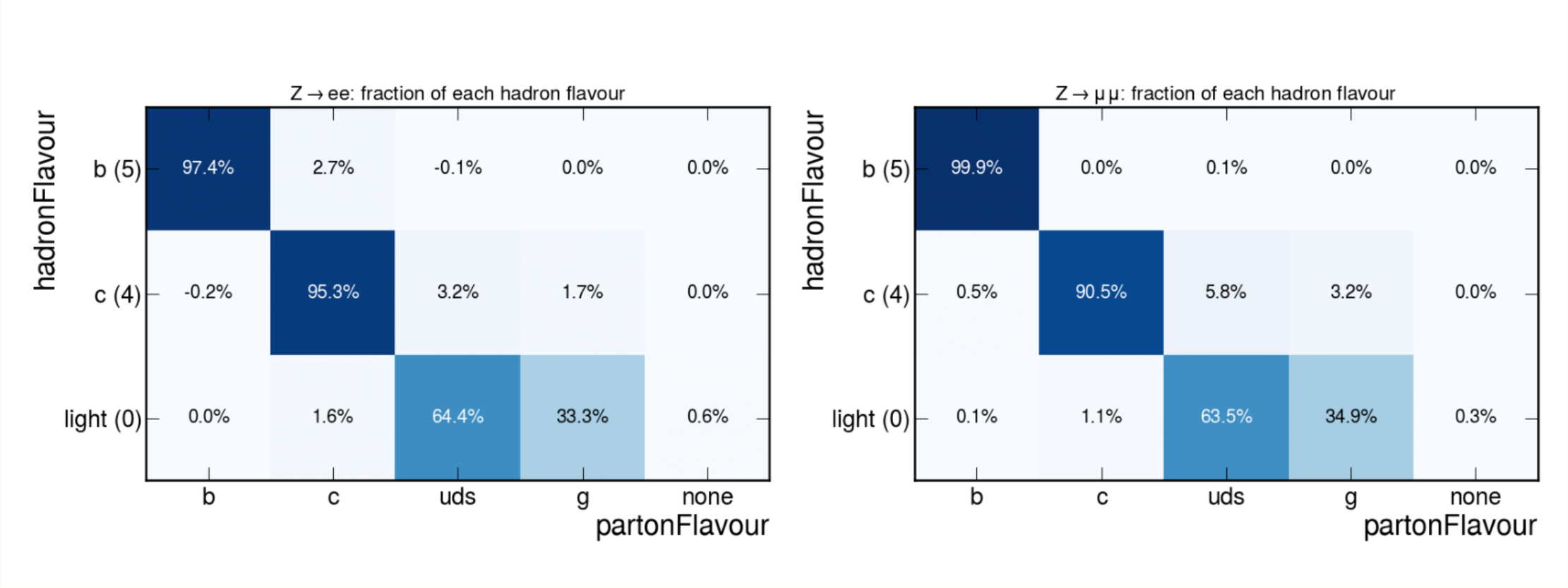
- Ghost-matched partons (last before hadronization) in the jet
- Hardest heavy parton wins: b (5), c (4), else hardest light parton
- Splits light jets into uds (1–3) and gluon (21); 0 = no parton matched

Fraction of scaled yield	hadron Z→ee	parton Z→ee	hadron Z→μμ	parton Z→μμ
b	0.042	0.041	0.045	0.047
c	0.106	0.116	0.083	0.085
light (hadron) / uds + g + none (parton)	0.852	0.843	0.872	0.868
of which g / none (parton)		0.286 / 0.005		0.307 / 0.003

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · leading AK4 jet, scaled to 7.98 fb^{-1} · 22

Hadron vs parton flavour migration

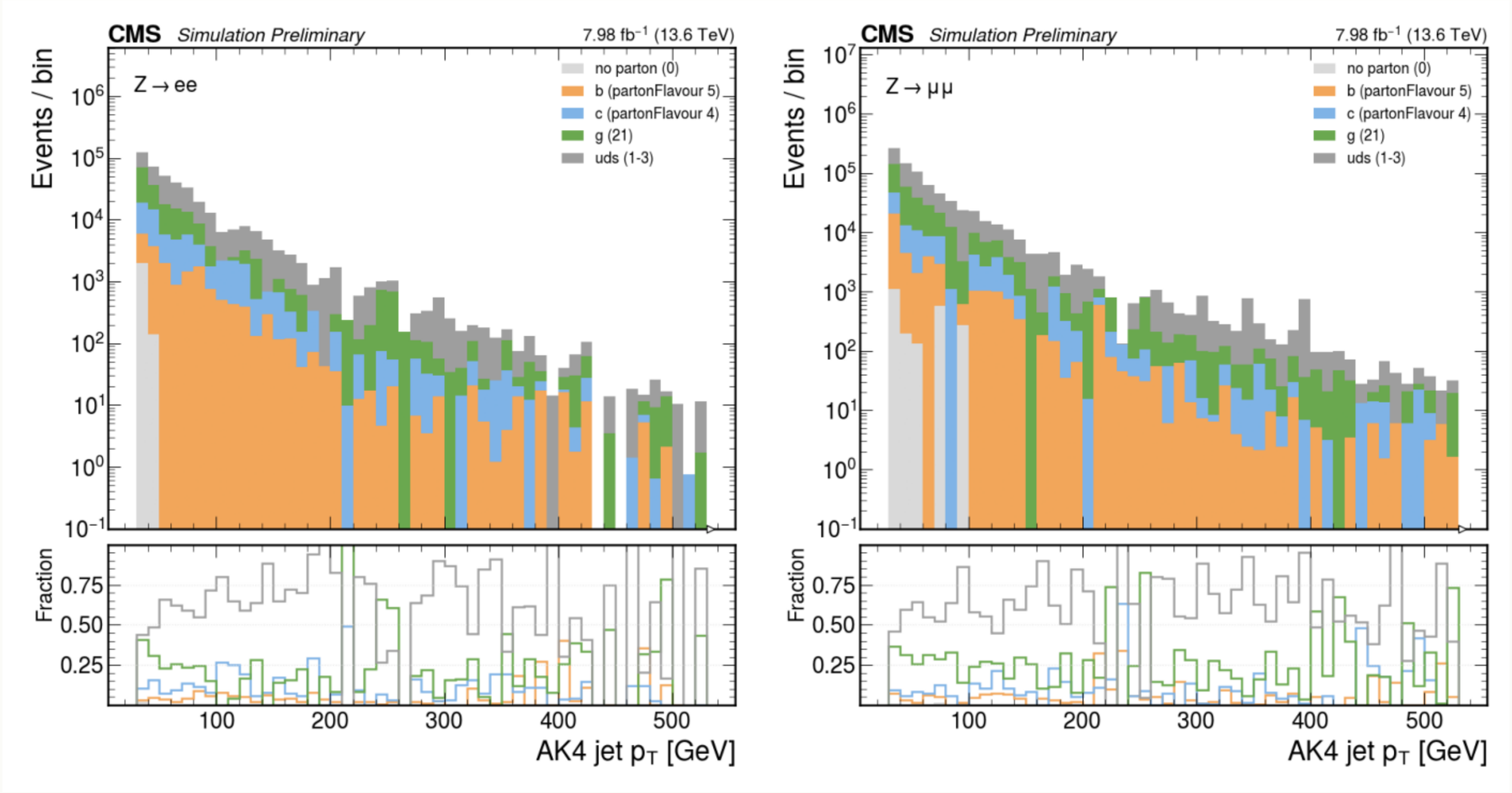
Each row shows where the jets of one hadronFlavour land in partonFlavour. 99% of hadron-b and 92% of hadron-c jets also have partonFlavour b / c (both channels). Hadron-light jets split into uds (64%) and gluon (34%); 1.4% carry a c or b parton without a matching hadron.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · rows: hadronFlavour, each sums to 100% (scaled) · small negative entries = NLO negative weigh

Leading AK4 jet pT, by partonFlavour

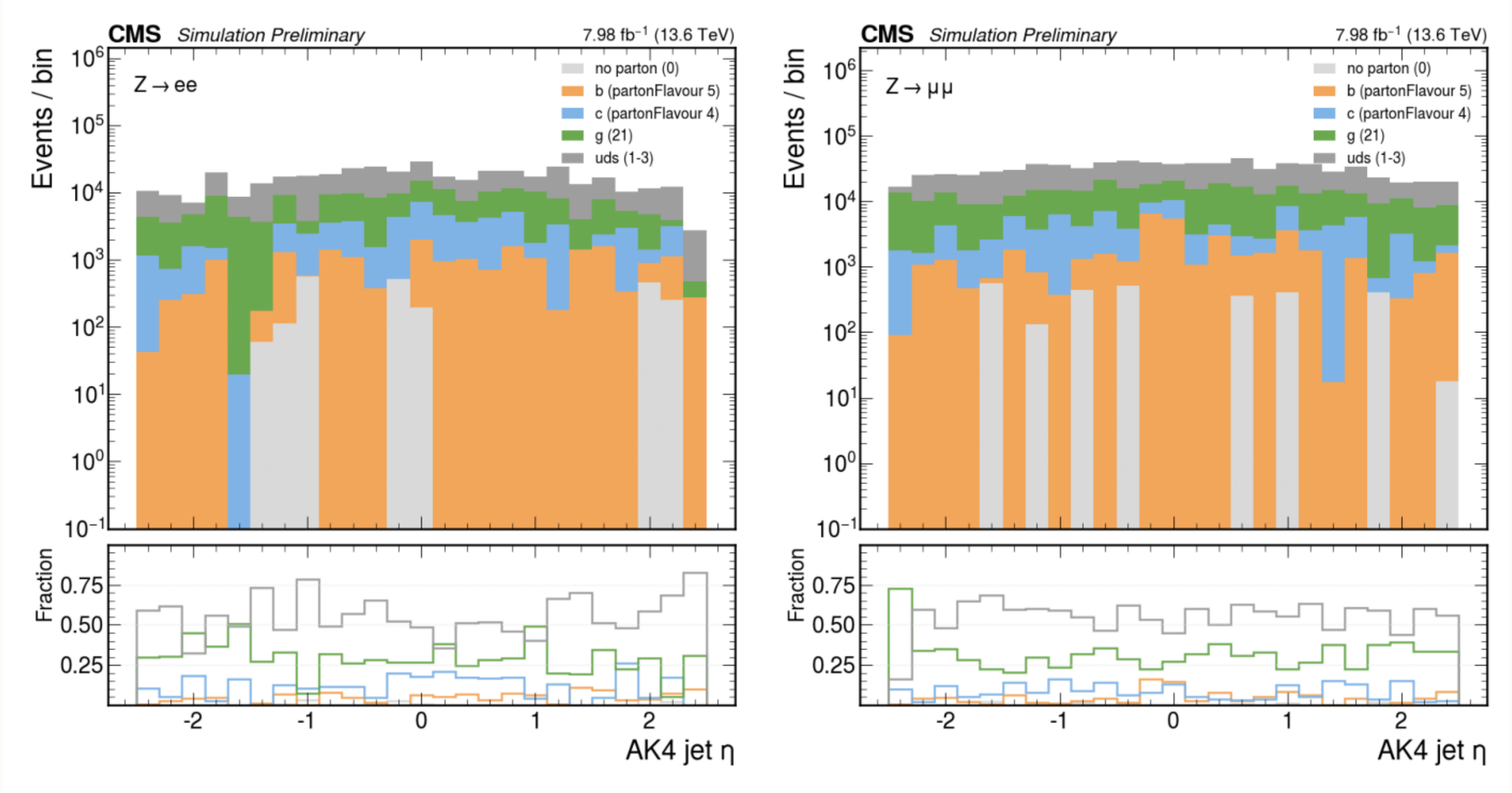
Same as page 13, stacked by |partonFlavour| of the leading jet.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 24

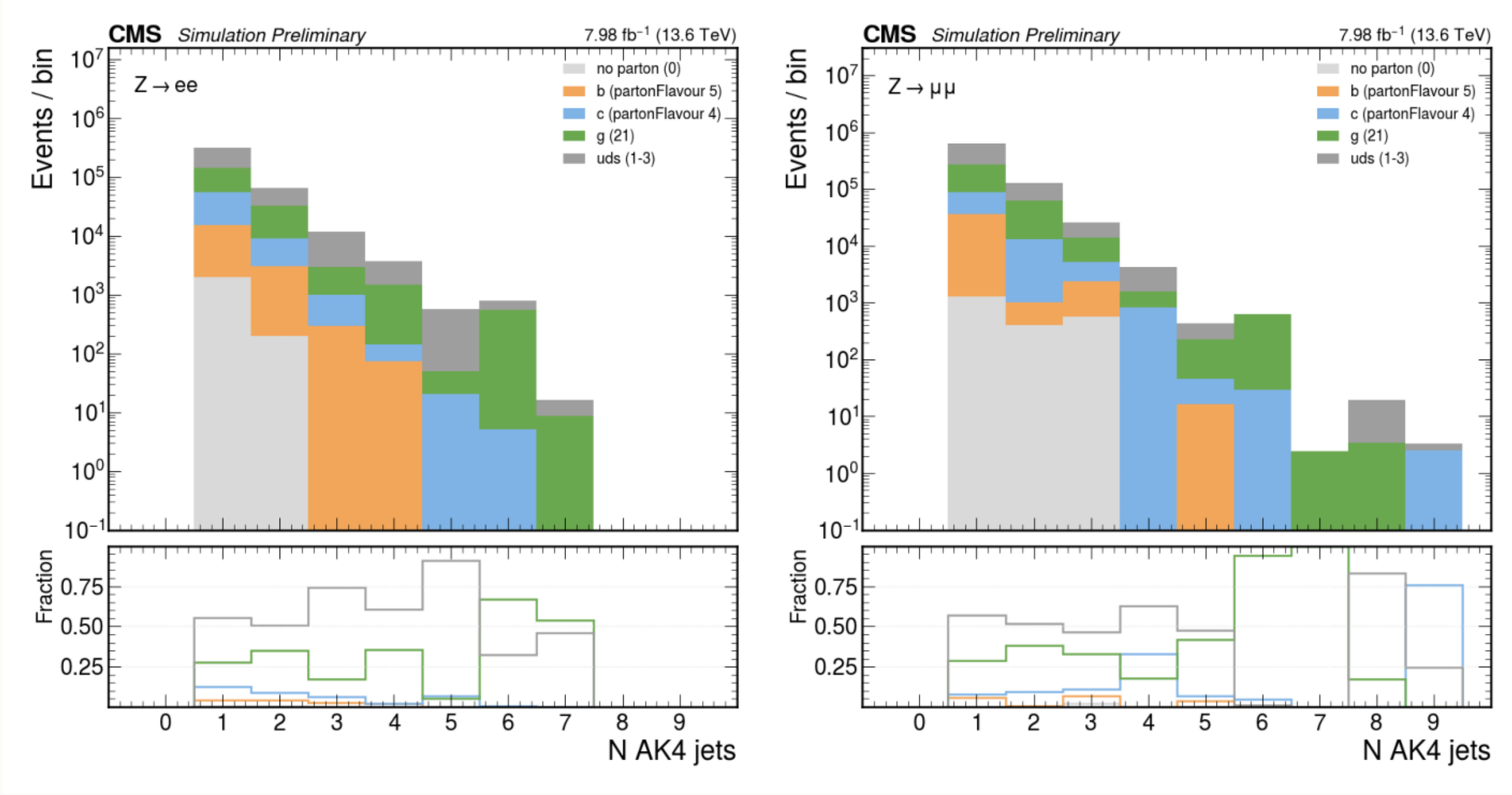
Leading AK4 jet η, by partonFlavour

Same as page 14, stacked by |partonFlavour| of the leading jet.



AK4 jet multiplicity, by partonFlavour

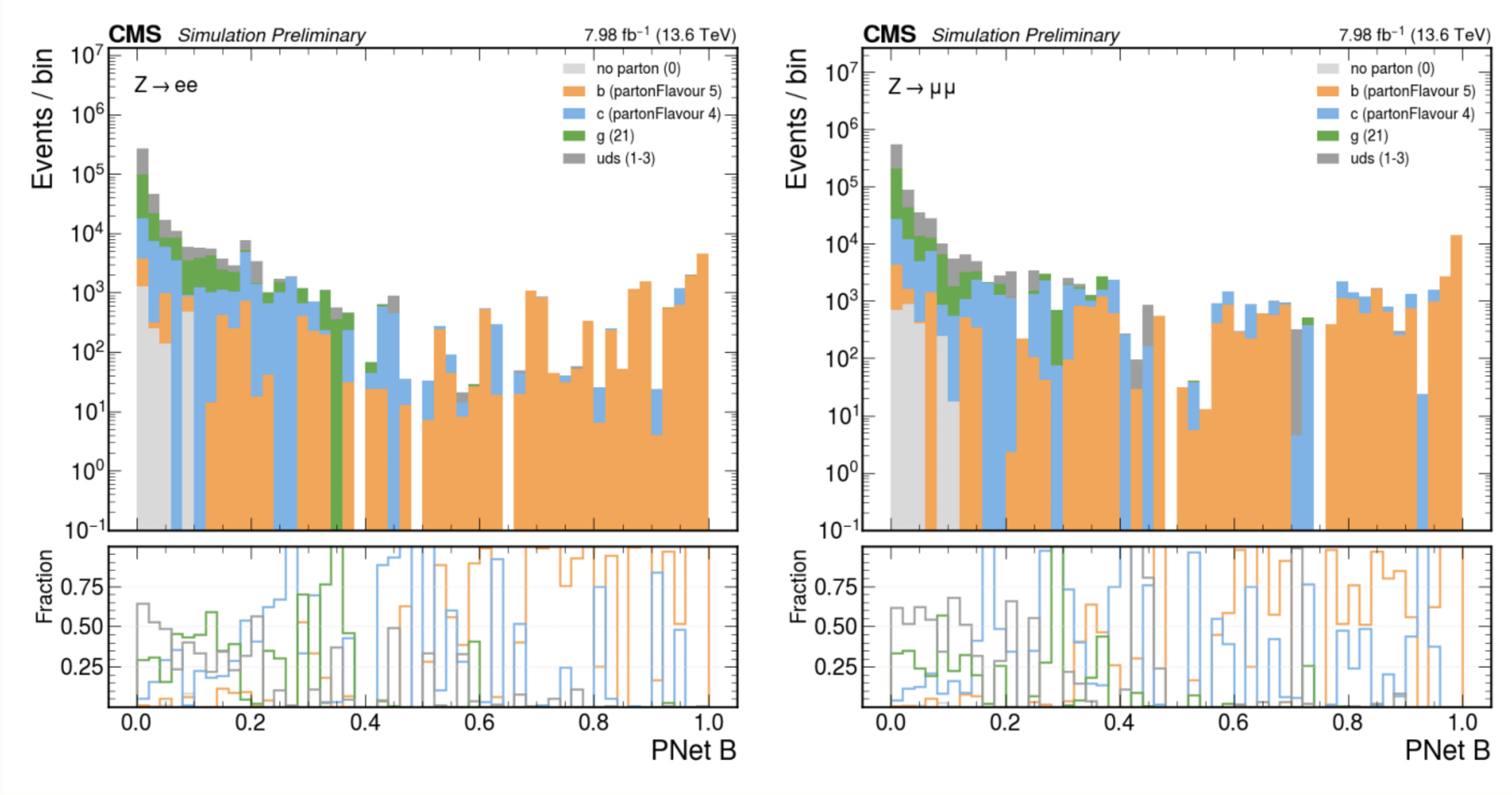
Same as page 15, stacked by |partonFlavour| of the leading jet.



Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 26

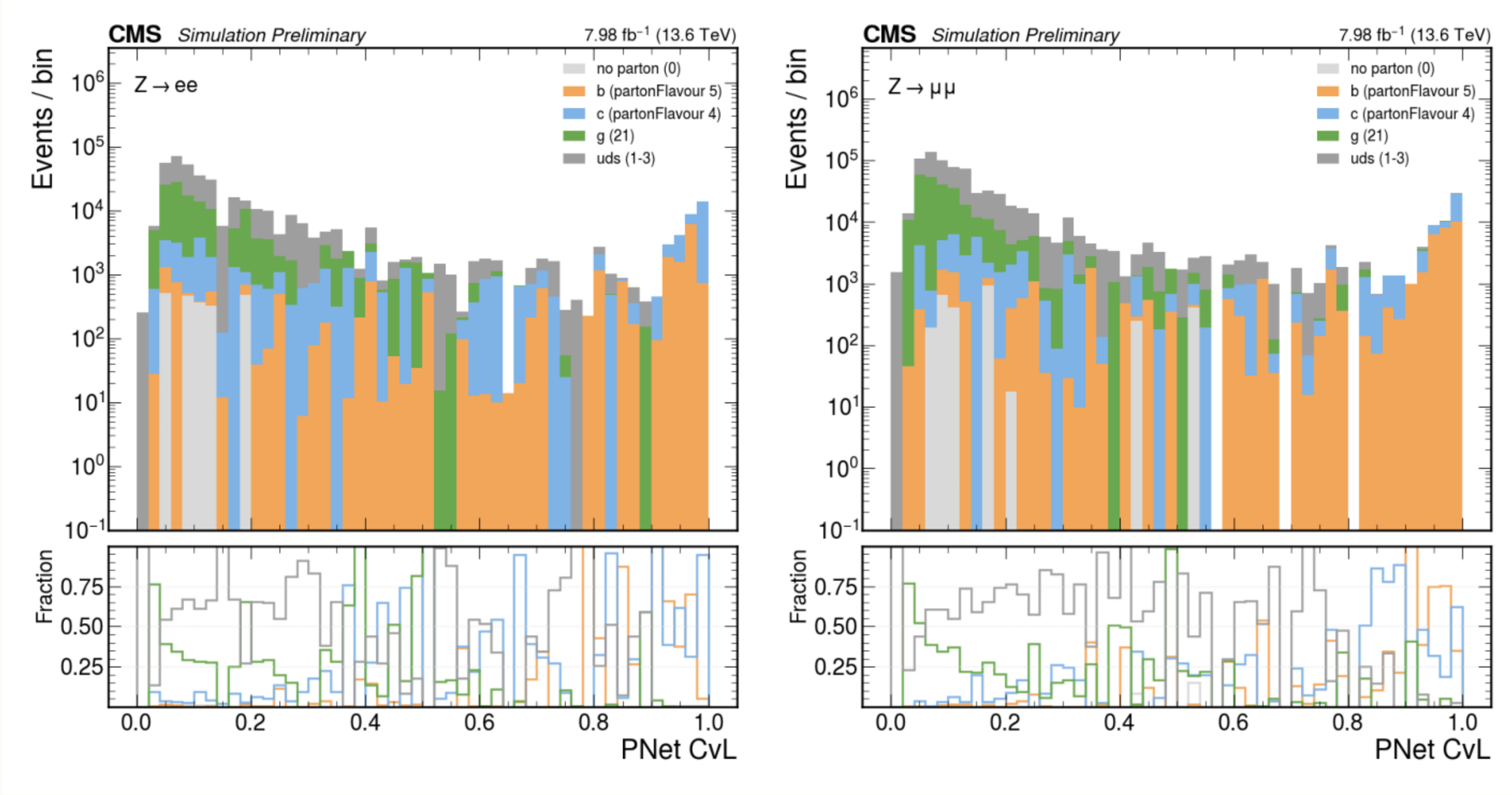
PNet B, by partonFlavour

Same as page 17, stacked by |partonFlavour| of the leading jet.



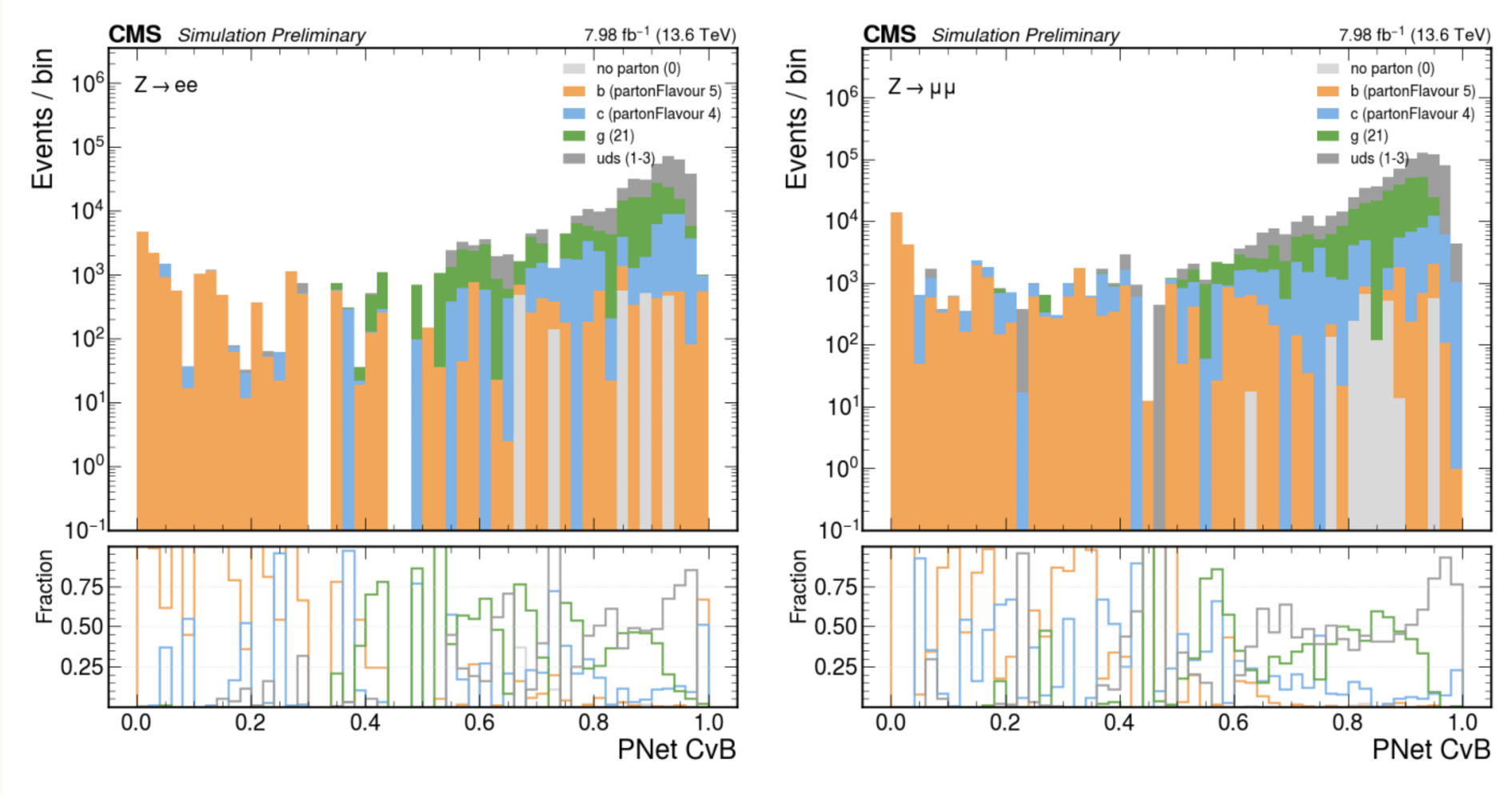
PNet CvL, by partonFlavour

Same as page 18, stacked by |partonFlavour| of the leading jet.



PNet CvB, by partonFlavour

Same as page 19, stacked by |partonFlavour| of the leading jet.



Leading jets with partonFlavour 0 are pileup

A jet with no GenJet (pT > 10 GeV) within ΔR < 0.4 (Jet_genJetIdx = −1) is a pileup jet. All 20 leading jets with partonFlavour 0 have no GenJet match (8 Z→ee, 12 Z→μμ).

Leading-jet partonFlavour	Z→ee no GenJet / all	Z→ee %	Z→μμ no GenJet / all	Z→μμ %
b	0 / 1,203	0.00	0 / 1,714	0.00
c	1 / 1,980	0.05	2 / 3,026	0.07
uds	7 / 12,991	0.05	4 / 20,170	0.02
g	20 / 5,185	0.39	37 / 8,184	0.45
none	8 / 8	100.00	12 / 12	100.00
all	36 / 21,367	0.17	55 / 33,106	0.17

- Not every pileup jet has partonFlavour 0: most unmatched leading jets carry a gluon label (a soft ISR/UE gluon is ghost-matched, but there is no GenJet).
- Pileup barely reaches the heavy-flavour yields: 0 of 2,842 hadron-b and 1 of 4,753 hadron-c leading jets are unmatched. Scaled, 1.9% (Z→ee) and 1.3% (Z→μμ) of leading jets are unmatched. This is above the raw fraction because the events come from the high-weight inclusive sample, so it waits for the full MC run.

N_jet bumps go away with more MC

20 files per sample instead of 1. Z→ee N_jet = 6: 801.9 (N_eff 1.8) → 45.5 (N_eff 0.7); Z→μμ N_jet = 6: 630.0 (N_eff 1.1) → 108.4 (N_eff 2.9), N_jet = 8: 19.7 (N_eff 1.4) → 0.0 (N_eff 0.0). Both spectra now fall smoothly up to N_jet = 7. The new Z→ee N_jet = 8 point, 33.7 (N_eff 1.1), is again a single inclusive-sample event: N_jet ≥ 6 still needs the full run.

